nature methods

SUPPLEMENTARY INFORMATION

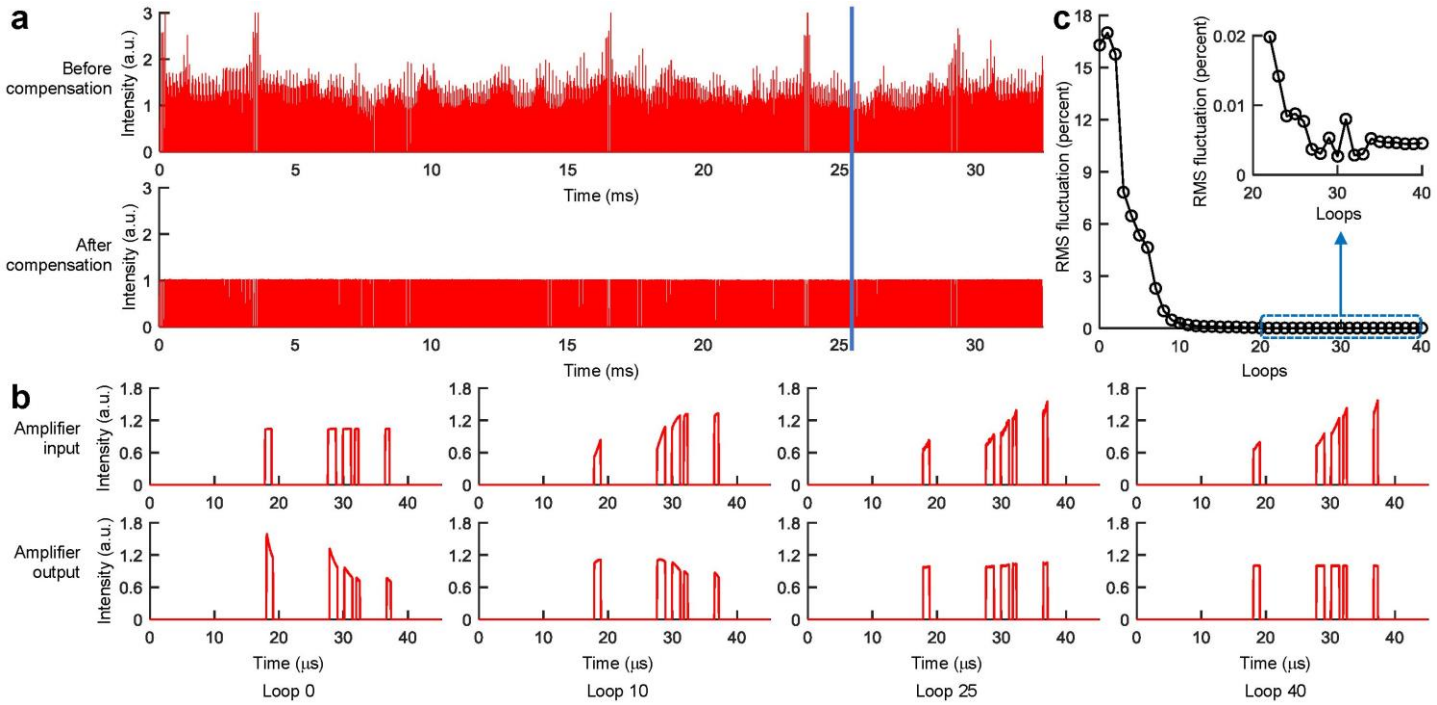
https://doi.org/10.1038/s41592-019-0663-9

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An adaptive excitation source for high-speed multiphoton microscopy

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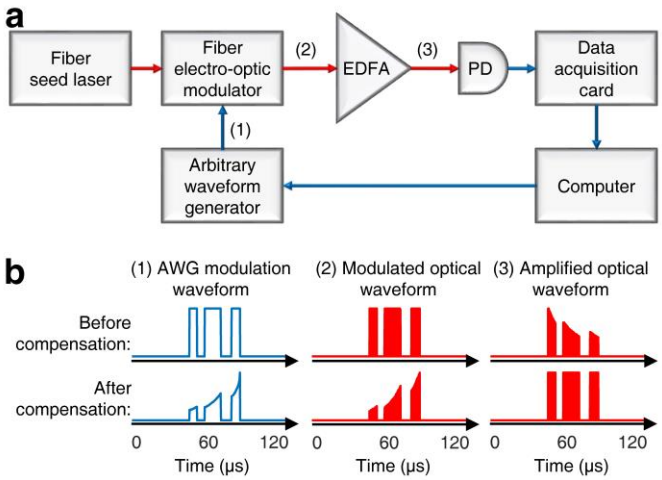
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Supplementary Figure 1

Demonstration of compensation of the gain transient in the EDFA.

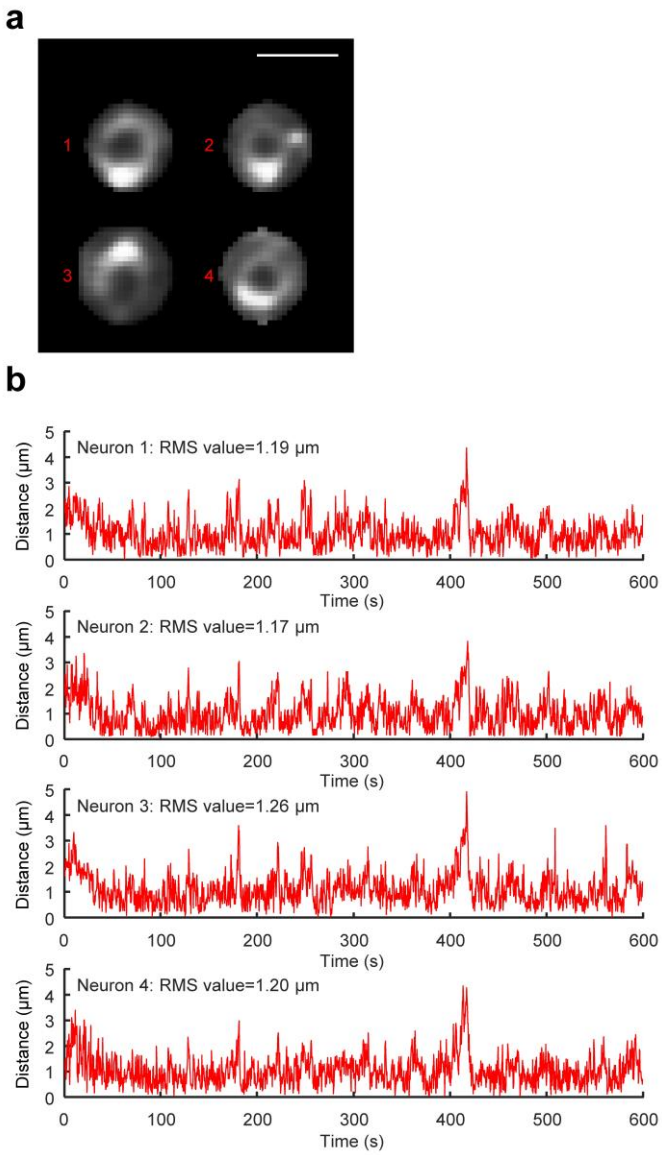
a, A comparison of the amplified pulses before and after compensation. The envelope of the peaks of the pulses is shown. The pulse sequences correspond to one frame (512x512 pixels) of activity recording where only the somas are illuminated. **b**, Zoomed-in view of the input and output of the EDFA for a small segment of time (the blue region in **a**) after various number of iterations (indicated at the bottom of each panel) using the feedback loop. **c**, RMS fluctuation of the pulse intensity as a function of the number of iterations using the feedback loop. The insert is a zoom-in view.



Supplementary Figure 2

The automatic feedback loop for compensation of the gain transient in the EDFA.

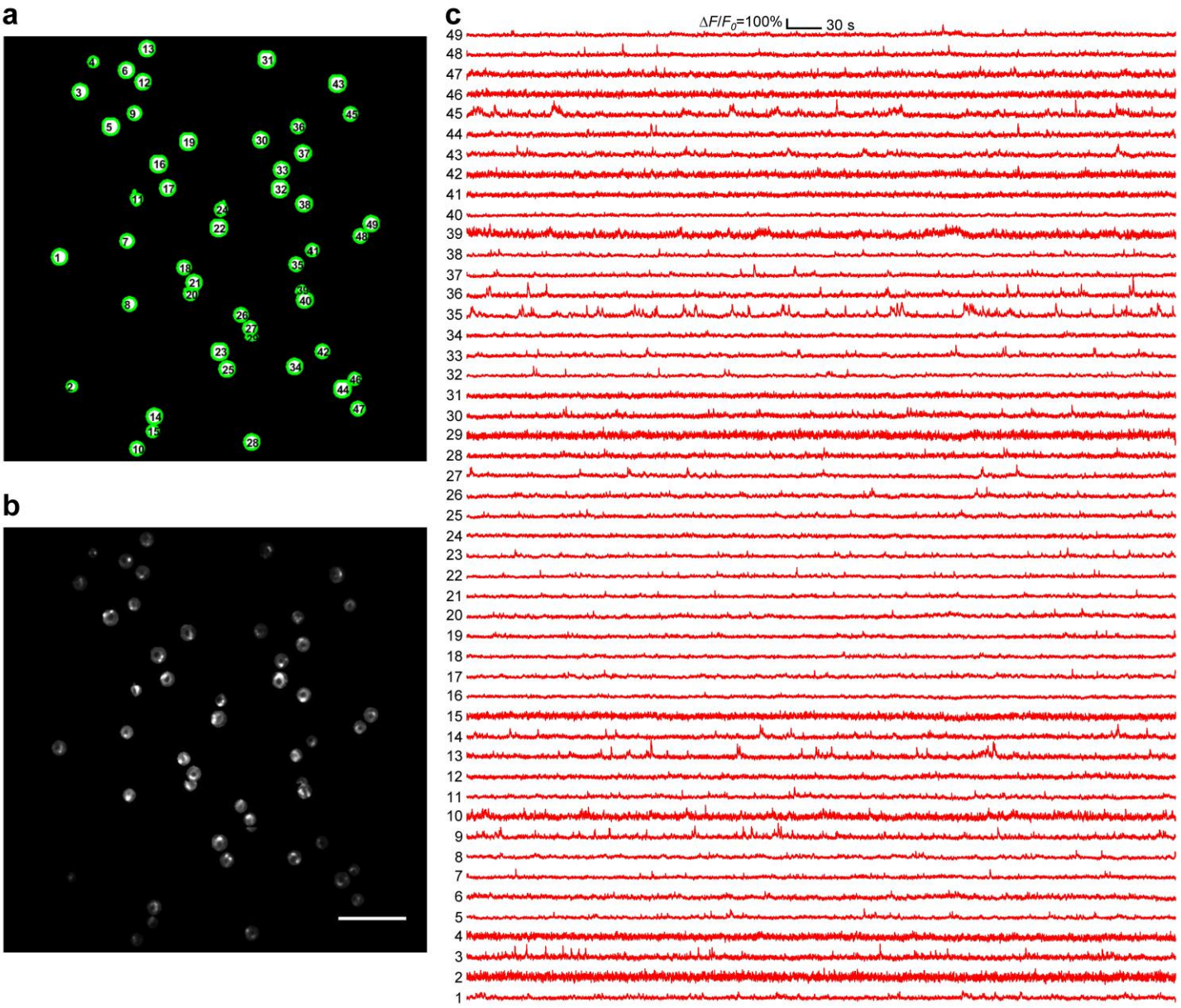
a, Diagram of the feedback loop. The electrical and optical paths are indicated in blue and red, respectively. **b**, The electrical (blue) and optical (red) waveforms at three positions indicated by (1), (2), and (3) in **a**. A detailed evolution of the waveforms is shown in Supplementary Video 1.



Supplementary Figure 3

Quantitative measurement of the neuron movement due to mouse motion.

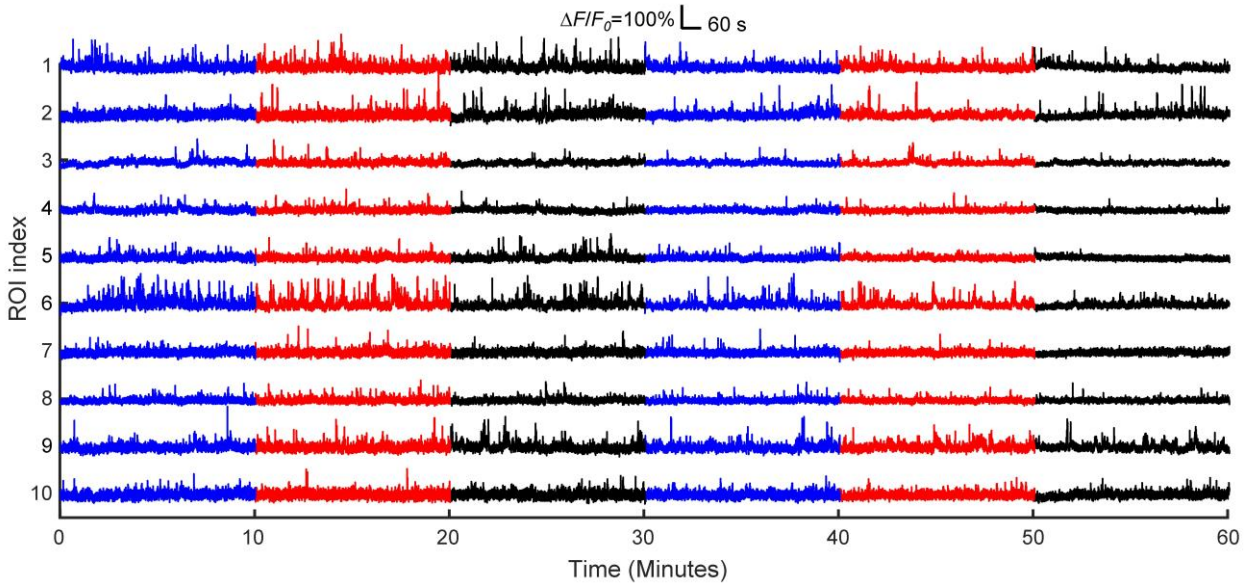
Four neurons in a recording session of an awake mouse (Figure 2) are shown in **a** and Supplementary video 2. Scale bar, 20 μm. **b**, The measured movement of the center of the neuron in each frame and the average movement for the whole recording session. The root-mean-square (RMS) value of the neuron movement is about 1.2 μm and the maximum movement is about 5 μm. Therefore, to capture the entire neuron, the radius of the ROI should be ~ 5 μm (or ~ 4 pixels for a FOV of 620x620 μm at 512x512 pixels) larger than the neuron.



Supplementary Figure 4

Spontaneous activity traces of all the neurons shown in Figure 2.

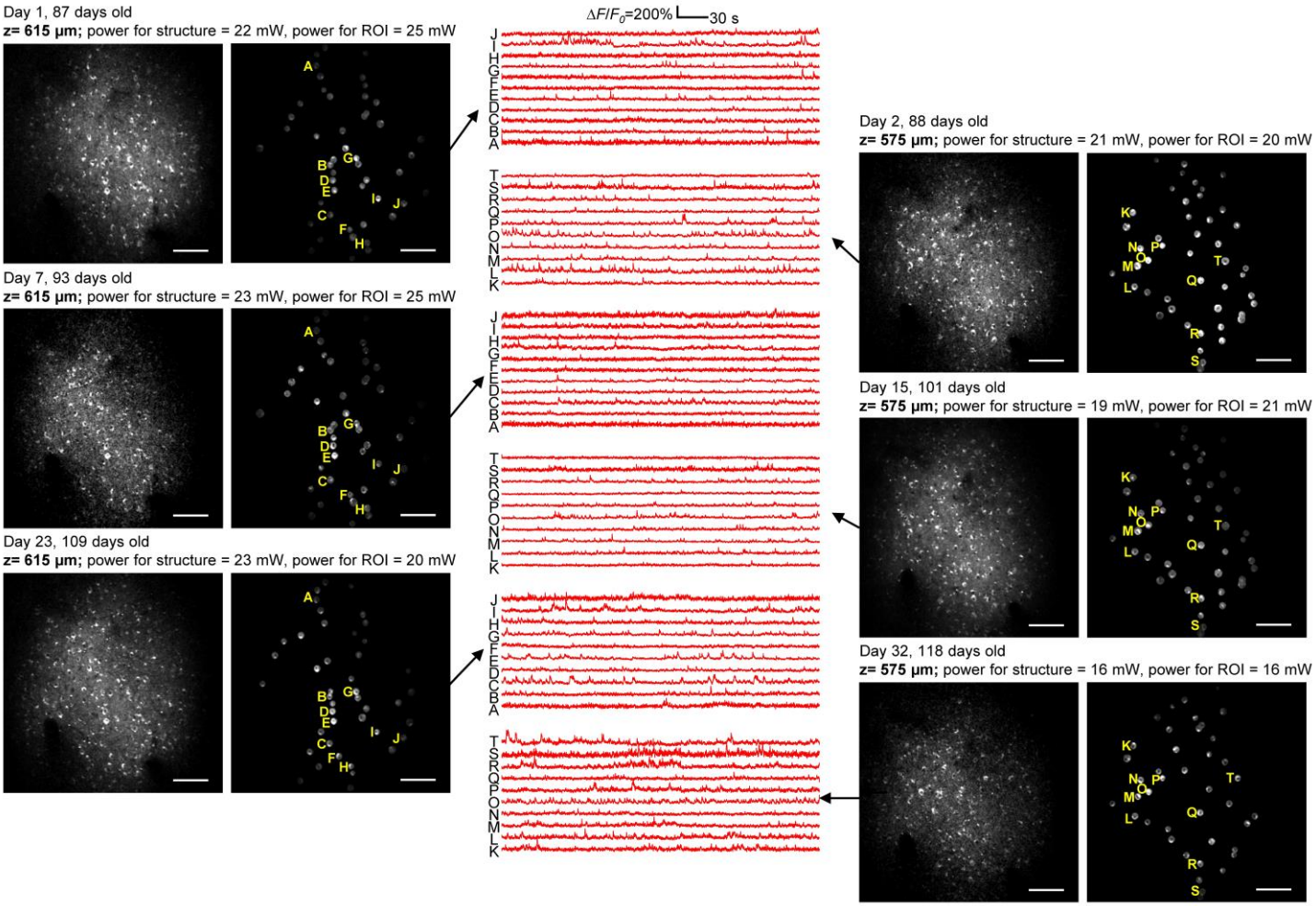
Neurons are indexed in **a**, and the corresponding ROIs are shown in **b**. Scale bar, 100 μm. **c**, Spontaneous activity traces recorded from the indexed neurons (1-49), at a frame rate of 30 Hz. To the left of each trace is the index of the neuron. All traces were normalized to each individual baseline.



Supplementary Figure 5

60-minute recording of spontaneous activity of jRGECO1a-labeled neurons shown in Fig. 2 with 3PM.

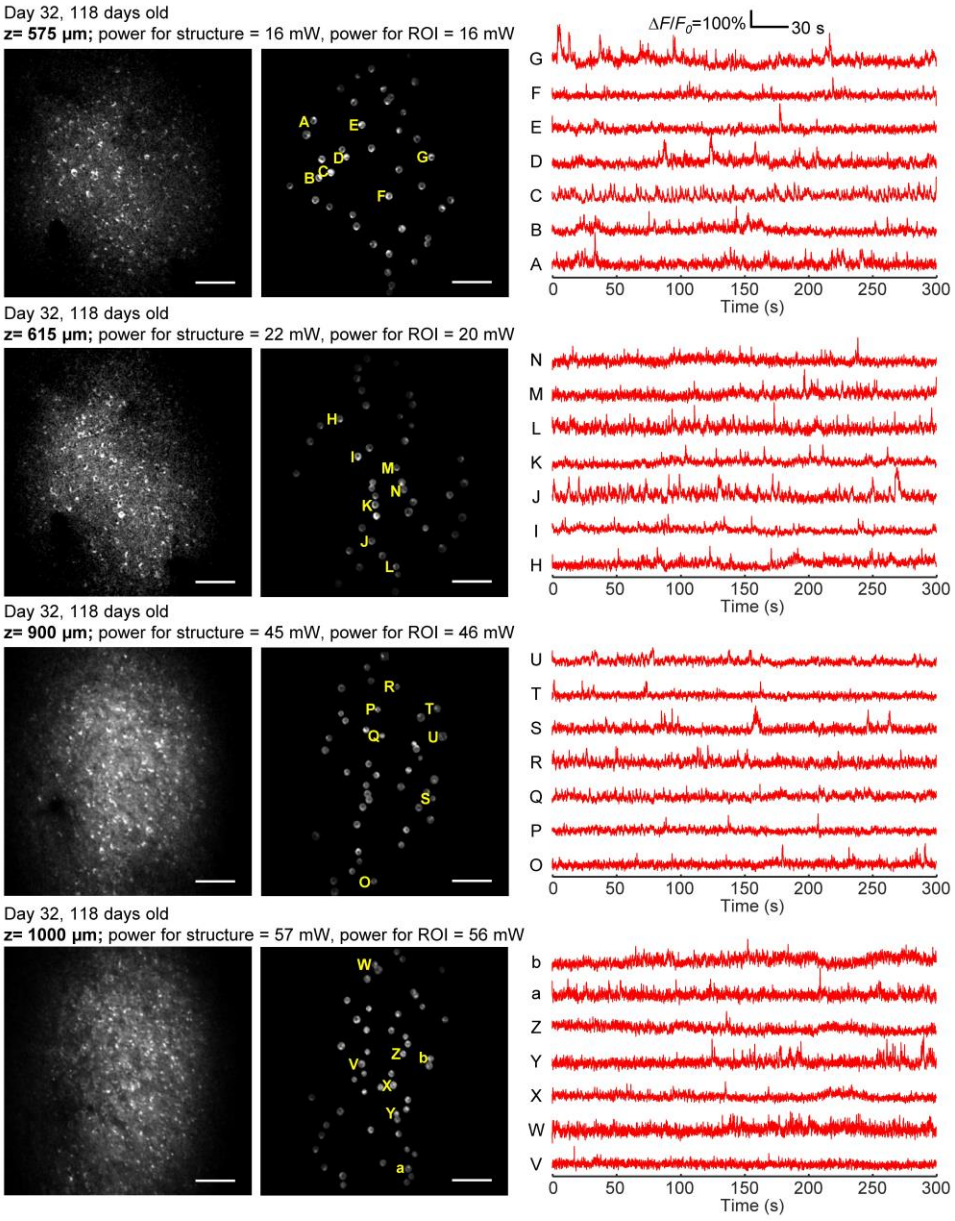
Six consecutive recording sessions are shown with different colors. The duration of each recording session is 10 minutes (18,000 frames), limited by the file size of the acquisition software. The time gap between two successive recording sessions is less than 5 seconds.



Supplementary Figure 6

Longitudinal recording of neuronal activity at the same locations in an awake animal over 4 weeks.

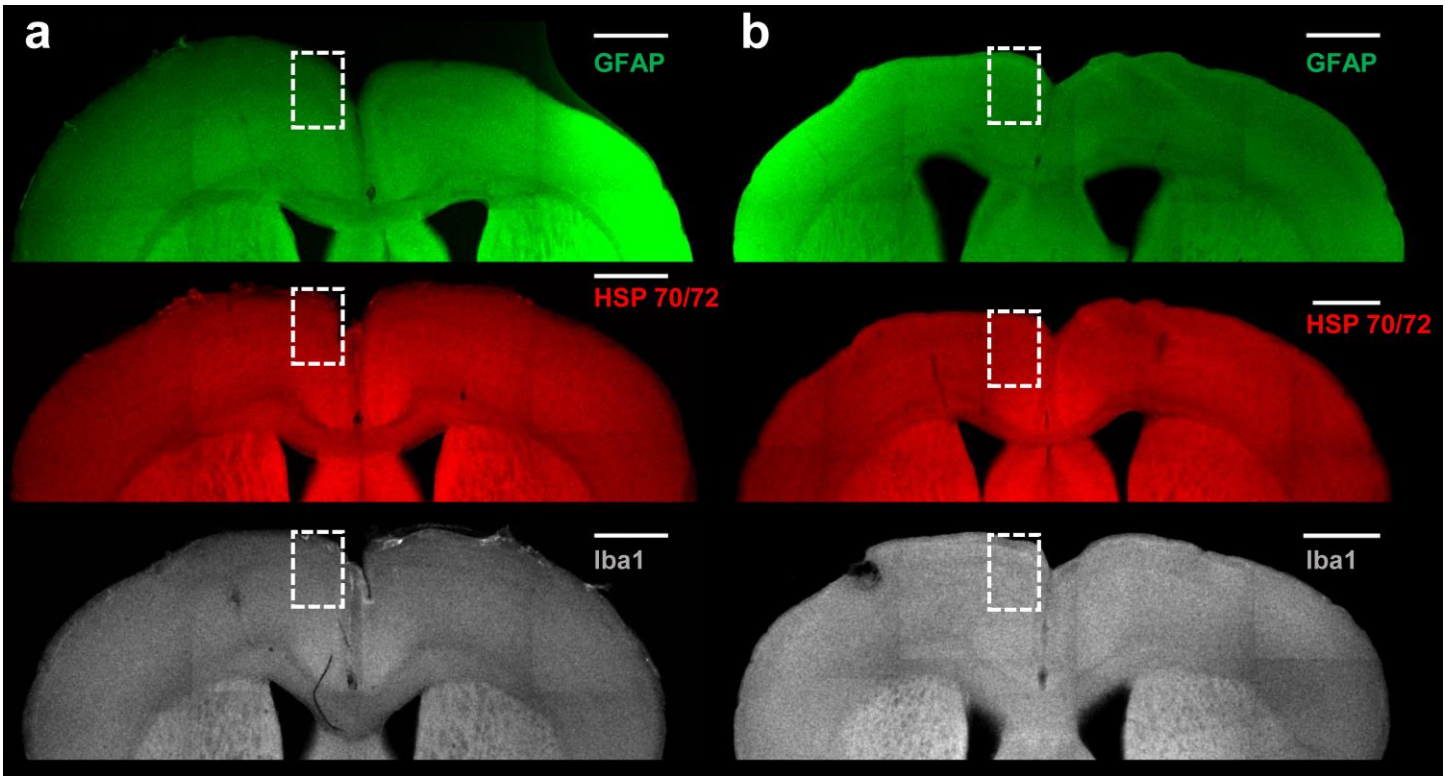
Spontaneous activity traces from cortical neurons were recorded in a jRGECO1a mouse under awake condition (12-17 weeks old, male) over a period of 4 weeks. High resolution structural and ROI images of two recording sites at 615 μm and 575 μm are shown in the left and right panels. The activity traces of a fraction of neurons are shown in the middle panels, recorded at 30 Hz frame rate, and each was normalized to its baseline. The neurons are indexed alphabetically, and the same unique letter for each neuron is used in all imaging sessions. The days after the first imaging session, the age of the mouse, the average power used for activity recording and the imaging depth are indicated at the top of each image. The images have a FOV of 620x620 μm with 512x512 pixels/frame. Scale bars, 100 μm.



Supplementary Figure 7

Recording of spontaneous activity of jRGECO1a-labeled neurons at various depths of the same mouse after performing the chronic imaging shown in Supplementary Figure 6.

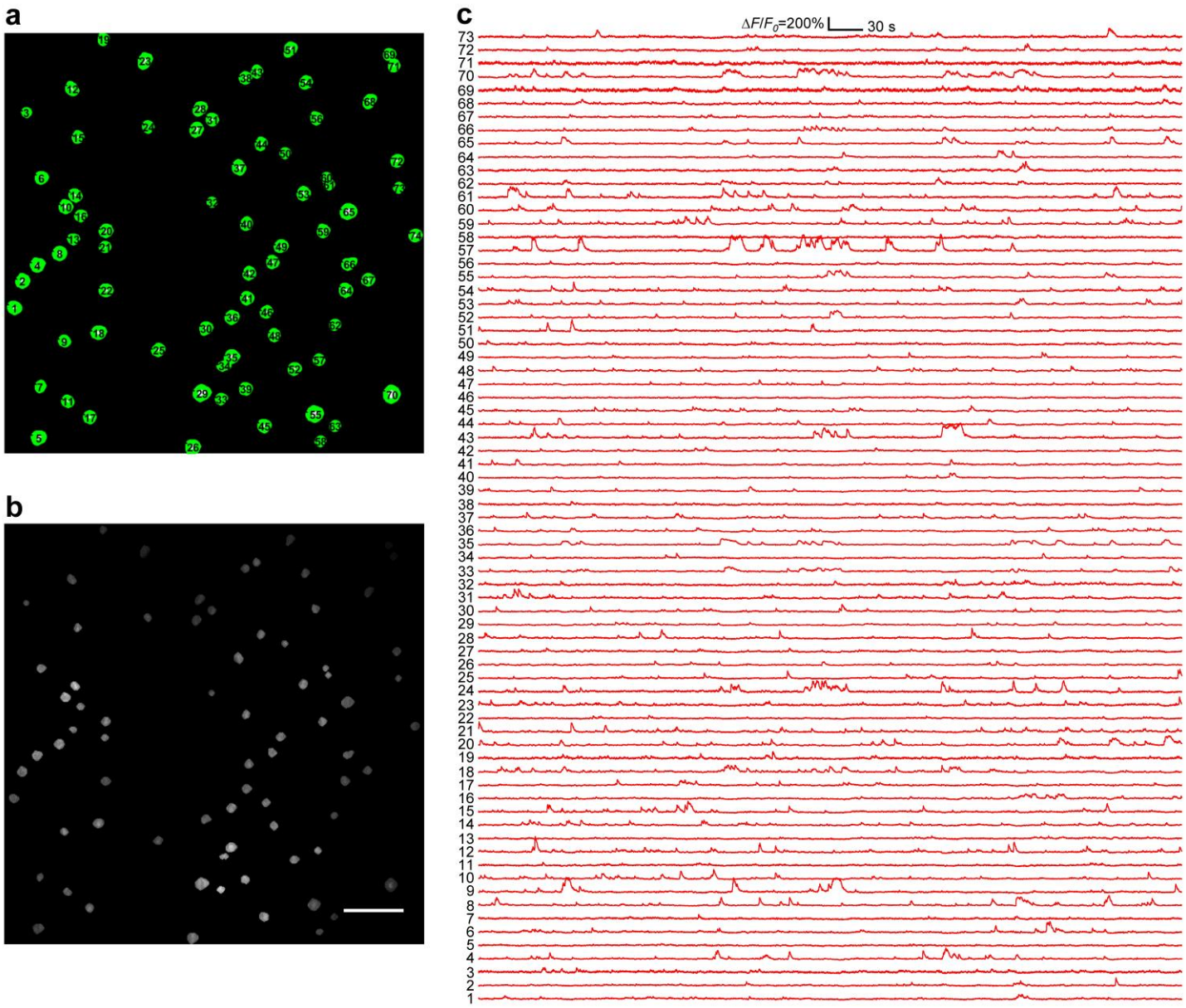
High resolution structural and ROI images of four recording sites at 575 μm, 615 μm, 900 μm and 1000 μm are shown in the left panels. The activity traces of a fraction of neurons are shown in the right panels, recorded at 30 Hz frame rate, and each was normalized to its baseline. The neurons are indexed alphabetically. The days after the first imaging session, the age of the mouse, the average power used for activity recording and the imaging depth are indicated at the top of each image. The images have a FOV of 620x620 μm with 512x512 pixels/frame. Damage assessment of the imaged regions using immunohistology is shown in Supplementary Figure 8(b). Scale bars, 100 μm.



Supplementary Figure 8

Immunohistology of the coronal sections of mouse brains after imaging.

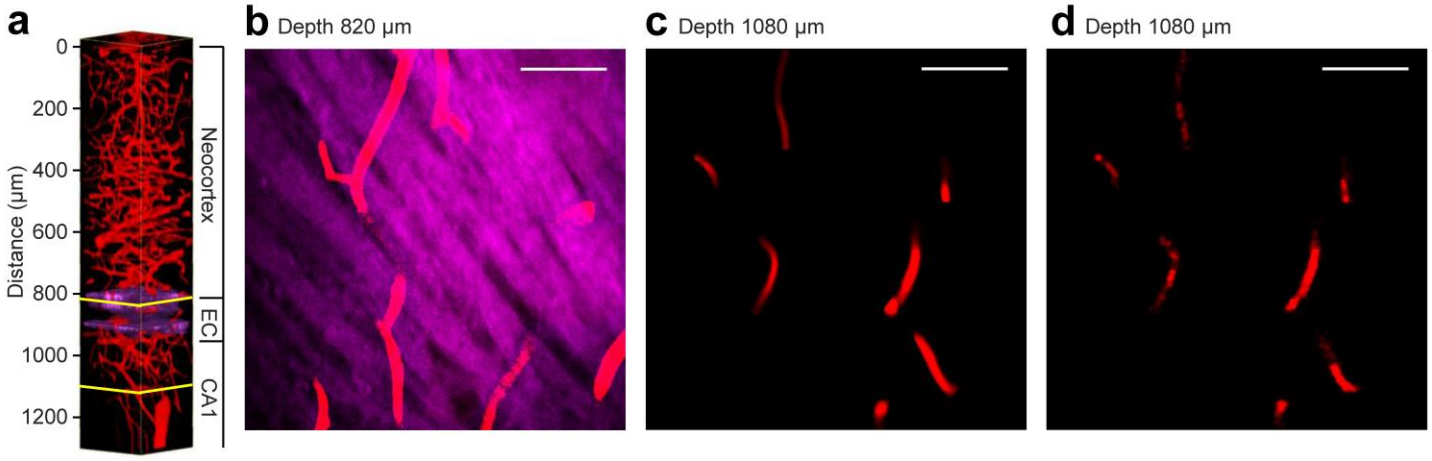
The imaging area is marked with dashed white rectangular box. **a**, The brain was imaged for a recording session of ~ 60 min (supplementary figure 5). **b**, The brain was imaged for 14 recording sessions over ~ 4 weeks (supplementary Figures 6 and 7). Scale bars, 1 mm.



Supplementary Figure 9

Spontaneous activity traces of the neurons shown in Figure 3.

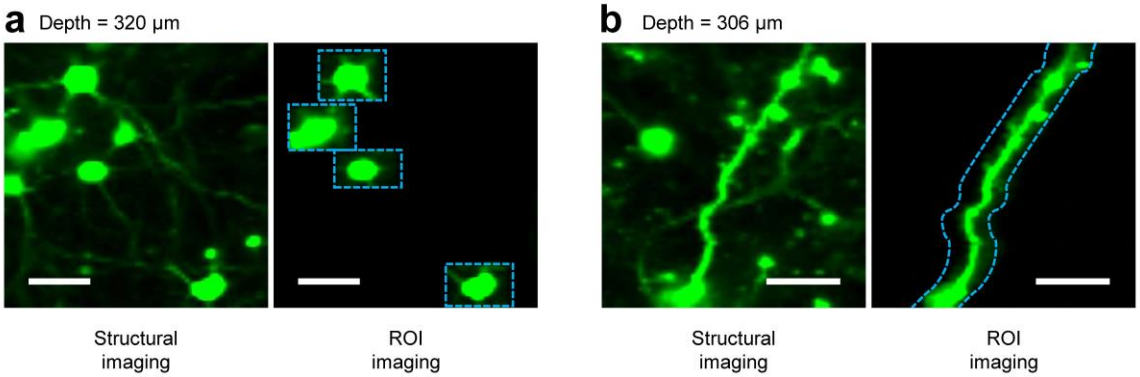
Neurons are indexed in **a**, and the corresponding ROIs are shown in **b**. Scale bar, 100 μm. **c**, Spontaneous activity traces recorded from the indexed neurons (1-73) at a frame rate of 30 Hz. To the left of each trace is the index of the neuron. All traces were normalized to each individual baseline.



Supplementary Figure 10

In vivo 3PM of the red blood cell flow in the capillaries in a mouse brain (3 months old, similar data for *n* = 2 mice).

Texas red dextran was injected to label blood vessels. **a**, Three-dimensional reconstruction of images of blood vessels in the mouse cortex and the hippocampus (red, blood vessels; magenta, third-harmonic generation (THG) of external capsule (EC)). **b**, A selected frame in the EC. **c**, A structural image of capillaries located at 1080 μm beneath the dura with a FOV of 220x220 μm, 512x512 pixels/frame, 30 frames/s, averaged over 100 seconds. **d**, Imaging of the capillaries (i.e., the ROIs) using the AES with the same FOV and frame rate but no averaging. Red blood cells can be seen as shadows flowing through the fluorescent plasma. Two recording sessions are shown in Supplementary Videos 6 and 7. Scale bars, 50 μm.



Supplementary Figure 11

In vivo imaging of neurons labeled with YFP in a mouse brain, using the AES and a commercial MPM (FV1000MPE, Olympus).

Structural and ROI imaging of neurons **a** and dendrites **b**. The blue dashed curves show the ROIs. The imaging depths are indicated at the top of the figures. Scale bars, 30 μm.

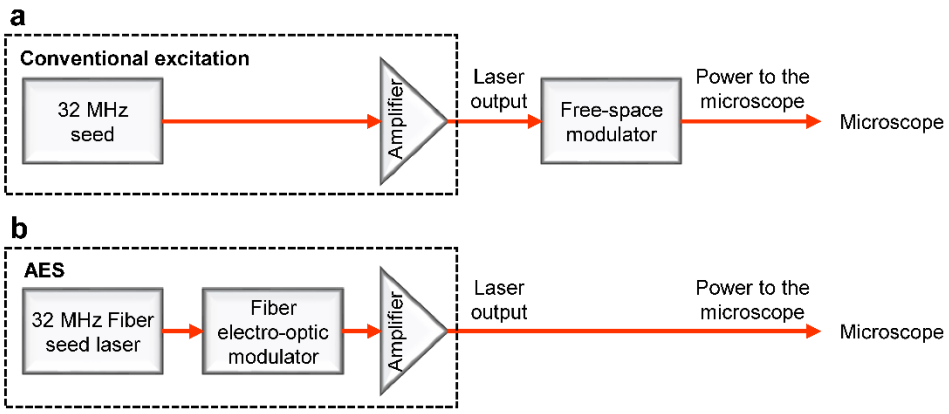
Supplementary Note 1:

Comparison of the AES to external modulation after the laser output

While modulation of laser power has been done routinely in the past, e.g., blanking the beam or enhancing the dynamic range, the modulation was always performed after the laser or amplifier output (i.e., outside the excitation source)[1,2], as shown in Supplementary Fig. SN1.1. Even though the same adaptive pulse pattern could be obtained by external free-space modulation after the laser output, the loss due to modulation is high, e.g., more than 30 times (based on our experimental results), without even counting the loss of the modulator itself. By losing ~ 97% of the laser output power, external modulation of the laser source requires 30X higher output power than the AES since all the AES output power is used for imaging. This makes external modulation of the laser impractical and sometimes impossible (see below) to achieve the desired imaging performance. In addition, by placing the modulator inside the excitation source, the AES allows a high-speed, low-power (only need to handle < 100 mW), and low-cost fiber-optic modulator to perform the intensity modulation.

For a conventional Ti:Sapphire laser with 80 MHz repetition rate to achieve the same 2PM imaging performance shown in Fig. 3, the required output power would be > 3 W using external free-space modulation, assuming the transmission of the microscope is 50% and the modulator itself is lossless. Such a power level (> 3 W at 920 nm) is beyond the reach of existing two-photon excitation sources.

For 3PM, existing optical parametric chirped-pulse amplification (OPCPA) system has a maximum repetition rate of ~ 5 MHz while being pumped at 50 to 100 W. To match the performance reported in this paper using external modulation, these OPCPA systems need to scale up to 32 MHz output repetition rate before being externally modulated, again assuming the modulator itself is lossless. Such a high repetition rate would necessitate pump power of at least 300 W, and is beyond the reach of existing three-photon excitation sources.



Supplementary Figure SN1.1

Comparison of an excitation source with external modulation (a) and an AES with internal modulation (b).

Supplementary Note 2:

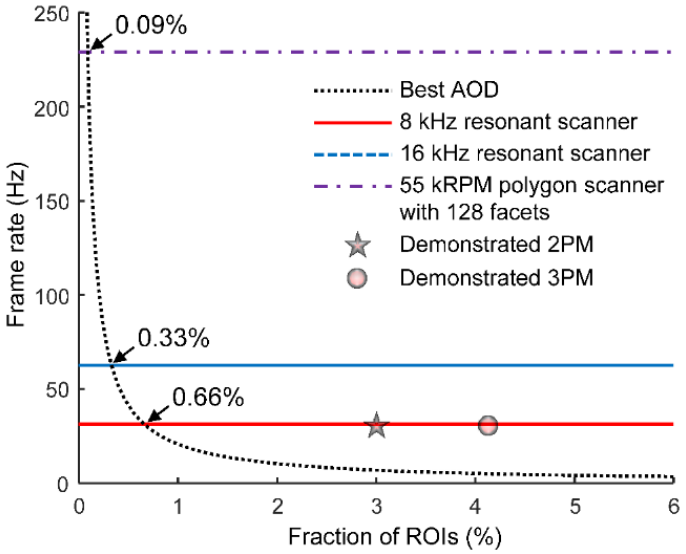
Comparison of AES and AOD microscopy

1. Comparison of AES and point-scanning RAMP

Both AES and point-scanning RAMP using AODs can perform ROI imaging (i.e., only the ROIs are illuminated by the excitation source), assuming that the laser is properly gated during the AOD transit time. A comparison of the frame rate is given in Supplementary Fig. SN2.1. The frame size is 512x512 pixels. For AES, the frame rate is ~ line frequency / 512, where the line frequency depends on the scanner. For point-scanning RAMP, the frame rate is approximately the AOD pixel rate / (512 * 512 * the fraction of ROIs) for ROI imaging. The fastest AOD scanner currently has a reported pixel rate of 54.3 kHz[3]. Supplementary Note 2 Figure 1 compares the best point-scanning AOD performance with several existing scanners that can be combined with the AES. For the AES with 8 kHz resonant scanner (e.g., CRS 8 kHz, Cambridge Technology), 16 kHz resonant scanner (e.g., SC-30, Electro-optical Products Corp.), and 55k RPM polygon scanner with 128 facets (e.g., SA24, Cambridge Technology), the frame rate is higher than point-scanning AOD when the fraction of ROIs is higher than 0.66%, 0.33% and 0.09%, respectively. Thus, imaging with the AES can significantly outperform the best point-scanning AOD system when the number of ROIs is large, such as the demonstrated 2PM and 3PM. Although the frame rate depends on the frame size (512x512 pixels), the same comparison holds for frames with different number of lines, e.g., 512x256 pixels.

2. Comparison of AES and line-scanning AOL microscopy

Multiphoton imaging using acousto-optic lens (AOL) for line scanning is another fast imaging method based on AODs[4]. The line rate of AOLs is similar to resonant scanners (e.g., ~ 20 kHz). Imaging a large of number of pixels per frame at a high frame rate has been achieved using the AOL microscope; however, the AOL microscope only performs ROI imaging in the direction perpendicular to the line scanning direction. By sacrificing the ability for ROI imaging along the scanning direction, AOL microscope reduces the power efficiency of excitation when multiple, spatially separated, ROIs appear along the same line. Therefore, while it overcomes the limit of the maximum pixel rate of a point-scanning RAMP, line-scanning AOL microscope is not as power-efficient as the ROI-imaging techniques such as the AES and point-scanning RAMP. In fact, the AES demonstrated here can be combined with AOL microscope to perform ROI imaging along the scanning line.



Supplementary Figure SN2.1

Comparison of the frame rate of AES with different scanners and the best AOD.

References

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